

Bryan M. Maitland

Curriculum Vitae
March 2026

Rocky Mountain Research Station,
USDA Forest Service, Boise, ID, USA
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Employment

2024–Present **Research Fish Biologist**, Rocky Mountain Research Station, USDA Forest Service
2023–2024 **Research Associate**, Dept. of Civil and Environmental Engineering, University of Wisconsin-Madison
2022–2023 **Postdoctoral Fellow**, Center for Limnology, University of Wisconsin-Madison
2020–2022 **WI Water Resources Postdoctoral Fellow**, Aquatic Sciences Center (UW-Madison) and WDNR

Education

2020 Ph.D. Ecology University of Wyoming
2015 M.Sc. Conservation Biology (Fisheries) University of Alberta
2011 B.S. Conservation Biology *cum laude* SUNY: College of Environmental Science and Forestry

Research Summary

- **Research Interests:** Fish; fisheries ecology and conservation; freshwater ecosystem responses to climate change and land use change; aquatic habitat connectivity; stable isotope ecology; spatial statistics and data science.
- **My research integrates** field ecology, isotopic and spatial analyses, and predictive modeling to understand how freshwater fish populations and food webs respond to environmental change.
- Since 2014 I have authored 25 papers in peer-reviewed journals (pages 1–2).
- On Google Scholar my h-index is 13 with 660 total citations (as of 13 March 2026).
- I have authored 1 R package as a result of my research with a total of 2187 downloads (page 2).
- I have contributed 47 papers and posters at AFS, SFS, ASLO, AGU, IAGLR, and regional chapter meetings (pages 6–8).

Peer-Reviewed Publications

* indicates shared first authorship. Underlined names indicate undergraduate research mentees.

25. Keppeler, F.W., Giarrizzo, T., Montaña, C.G., Adite, A., Akın, Ş., Angelini, R., **Maitland, B.M.**, Arantes, C.C., Benedito, E., Bokhutlo, T., El-Sabaawi, R., Garcia, A., Gonzalez-Bergonzoni, I., Hoeinghaus, D., Hoffman, J., ... Winemiller, K.O. (2026). What determines trophic niche breadth? A global analysis of freshwater fishes using isospaces. *Reviews in Fish Biology and Fisheries*, 36(13). <https://doi.org/10.1007/s11160-025-10023-2>
24. Sparks*, M.M., **Maitland***, **B.M.**, Felts, E., Swartz, A., & Frater, P. (2026). hatchR: A toolset to predict when fish hatch and emerge. *Fisheries*, 51(2), 49–59. <https://doi.org/10.1093/fshmag/vuaf078>
23. Barrus, N.T., **Maitland, B.M.**, & Rahel, F.J. (2024). Assessing a standardized method to identify optimal baselines for trophic position estimation in stable isotope studies of stream ecosystems. *Hydrobiologia*, 851, 4673–4691. <https://doi.org/10.1007/s10750-024-05618-y>
22. **Maitland, B.M.**, Bootsma, H.A., Bronte, C.R., Bunnell, D.B., Feiner, Z.S., Fenske, K.H., Fetzer, W.W., Foley, C.J., Gerig, B.S., Happel, A., Höök, T.O., Keppeler, F.W., Kornis, M.S., Lepak, R.F., McNaught, A.S., ... Jensen, O.P. (2024). Testing food web theory in a large lake: The role of body size in habitat coupling in Lake Michigan. *Ecology*, 105(10), e4413. <https://doi.org/10.1002/ecy.4413>
21. **Maitland, B.M.**, & Rahel, F.J. (2023). Aquatic food web expansion and trophic redundancy along the Rocky Mountain—Great Plains ecotone. *Ecology*, 104(7), e4103. <https://doi.org/10.1002/ecy.4103>
20. Ward, N.K., Lynch, A.J., Beever, E.A., Booker, J., Bouska, K.L., Embke, H., Houser, J.N., Kocik, J.F., Kocik, J., Lawrence, D.J., Lemon, M.G., Limpinsel, D., Magee, M.R., **Maitland, B.M.**, McKenna, O., ... Wilkening, J.L. (2023). Reimagining large river management using the Resist–Accept–Direct (RAD) framework in the Upper Mississippi River. *Ecological Processes*, 12(1), 48. <https://doi.org/10.1186/s13717-023-00460-x>
19. Feiner, Z.S., Shultz, A.D., Sass, G.G., Trudeau, A., Mitro, M.G., Dassow, C.J., Latzka, A.W., Isermann, D.A., **Maitland, B.M.**, Homola, J.J., Embke, H.S., & Preul, M. (2022). Resist-accept-direct (RAD) considerations for climate change adaptation in fisheries: The Wisconsin experience. *Fisheries Management and Ecology*, 29(4), 346–363. <https://doi.org/10.1111/fme.12549>
18. Kirk, M.A., **Maitland, B.M.**, Hickerson, B.T., Walters, A.W., & Rahel, F.J. (2022). Climatic drivers and ecological impacts of a rapid range expansion by non-native smallmouth bass. *Biological Invasions*, 24, 1311–1326. <https://doi.org/10.1007/s10530-021-02724-z>
17. Lapidés*, D.A., **Maitland***, **B.M.**, Zipper, S.C., Latzka, A.W., Pruitt, A., & Greve, R. (2022). Advancing environmental flows approaches to streamflow depletion management. *Journal of Hydrology*, 607, 127447. <https://doi.org/10.1016/j.jhydrol.2022.127447>
16. **Maitland, B.M.**, & Latzka, A.W. (2022). Shifting climate conditions affect recruitment in Midwestern stream trout, but depend on seasonal and spatial context. *Ecosphere*, 13(12), e4308. <https://doi.org/10.1002/ecs2.4308>

15. Medinski, N.A., **Maitland, B.M.**, Jardine, T.D., Drake, D.A.R., & Poesch, M.S. (2022). A catastrophic coal mine spill in the Athabasca River watershed induces isotopic niche shifts in stream biota including an endangered rainbow trout ecotype. *Canadian Journal of Fisheries and Aquatic Sciences*, 1–14. <https://doi.org/10.1139/cjfas-2021-0112>
14. **Maitland, B.M.**, Martínez del Río, C., & Rahel, F.J. (2021). Effect of temperature on ^{13}C and ^{15}N incorporation rates and discrimination factors in two North American fishes. *Canadian Journal of Fisheries and Aquatic Sciences*, 78(12), 1833–1840. <https://doi.org/10.1139/cjfas-2021-0057>
13. **Maitland, B.M.**, & Rahel, F.J. (2021). Nonlethal fin sampling of North American freshwater fishes for food web studies using stable isotopes. *North American Journal of Fisheries Management*, 41(2), 410–420. <https://doi.org/10.1002/nafm.10539>
12. Voter, C.B., Guerrero-Bolaño, F.J., Latzka, A.W., **Maitland, B.M.**, & Hauxwell, J.A. (2021). Adaptable university-agency early-career fellowship program creates a win-win-win for wisconsin's waters. *Journal of Contemporary Water Research and Education*, 174, 140–155.
11. Kirk, M.A., **Maitland, B.M.**, & Rahel, F.J. (2020). Spatial scale, reservoirs and nonnative species influence the homogenization and differentiation of Great Plains — Rocky Mountain fish faunas. *Hydrobiologia*, 847, 1–15. <https://doi.org/10.1007/s10750-019-04129-5>
10. Alston, J.M., **Maitland, B.M.**, Brito, B.T., Esmaeili, S., Ford, A.T., Hays, B., Jesmer, B.R., Molina, F.J., & Goheen, J.R. (2019). Reciprocity in restoration ecology: When might large carnivore reintroduction restore ecosystems? *Biological Conservation*, 234, 82–89. <https://doi.org/10.1016/j.biocon.2019.03.021>
9. Hickerson, B.T., **Maitland, B.M.**, & Walters, A.W. (2019). Effects of multiple nonnative fish on an imperiled cyprinid, Hornyhead Chub *Nocomis biguttatus*. *Transactions of the American Fisheries Society*, 148(6), 1132–1145. <https://doi.org/10.1002/tafs.10203>
8. **Maitland, B.M.**, O'Malley, B.P., & Stewart, D.J. (2019). Subsurface water piping transports plankton and prevents meromixis in a deep volcanic crater lake (Dominica, West Indies). *Hydrobiologia*, 839(1), 119–130. <https://doi.org/10.1007/s10750-019-04000-7>
7. Walker, R.H., **Maitland, B.M.**, LaSharr, T.N., Rosing, M.N., & Ben-David, M. (2018). Fate of juvenile salmonids stranded in off-channel pools: Implications for nutrient transfers. *Aquatic Sciences*, 80(1), 10. <https://doi.org/10.1007/s00027-017-0562-z>
6. Craig, L.S., Olden, J.D., Arthington, A.H., Entekhabi, S., Hawkins, C.P., Kelly, J.J., Kennedy, T.A., **Maitland, B.M.**, Rosi, E.J., Roy, A.H., Strayer, D.L., Tank, J.L., West, A.O., & Wooten, M.S. (2017). Meeting the challenge of interacting threats in freshwater ecosystems: A call to scientists and managers. *Elementa: Science of the Anthropocene*, 5(72), 1–15. <https://doi.org/10.1525/elementa.256>
5. Pandit, S.N., **Maitland, B.M.**, Pandit, L.K., Poesch, M.S., & Enders, E.C. (2017). Climate change risks, extinction debt, and conservation implications for a threatened freshwater fish: Carmine shiner (*Notropis percobromus*). *Science of the Total Environment*, 598, 1–11. <https://doi.org/10.1016/j.scitotenv.2017.03.228>
4. **Maitland, B.M.**, Poesch, M., & Anderson, A.E. (2016a). Prioritising culvert removals to restore habitat for at-risk salmonids in the boreal forest. *Fisheries Management and Ecology*, 23(6), 489–502. <https://doi.org/10.1111/fme.12188>
3. **Maitland, B.M.**, Poesch, M., Anderson, A.E., & Pandit, S.N. (2016b). Industrial road crossings drive changes in community structure and instream habitat for freshwater fishes in the boreal forest. *Freshwater Biology*, 61, 1–18.
2. **Maitland, B.M.**, Cooke, S.J., & Poesch, M. (2015). Finding the path to a successful graduate and research career: Advice for early career researchers. *Fisheries*, 40(8), 399–403. <https://doi.org/10.1080/03632415.2015.1065253>
1. O'Malley, B.P., & **Maitland, B.M.**, (2014). Observations on the freshwater zooplankton of Dominica, West Indies, including records from the former Matthieu landslide-dam lake. *Pan-American Journal of Aquatic Sciences*, 9(3), 228–333.

R-packages

1. **Maitland, B.M.**, Sparks, M.M., Felts, E., Swartz, A., & Frater, P.N. (2025). *hatchR: Predict Fish Hatch and Emergence Timing*. <https://github.com/bmait101/hatchR>

Technical reports

3. **Maitland, B.M.**, & Loheide, S.P. (2025). *Pumped storage hydropower in California: Environmental effects and opportunities for co-benefits*. University of Wisconsin-Madison.
2. Mitro, M., Feiner, Z.S., Latzka, A.W., Embke, H.S., Homola, J.J., Isermann, D.A., **Maitland, B.M.**, Sass, G.G., Shaw, S., Shultz, A.D., Tsyehaye, I., & Vander Zanden, J. (2021). *Wisconsin Initiative on Climate Change Impacts: Fisheries Working Group Report*. Nelson Institute for Environmental Studies, University of Wisconsin-Madison.
1. Roseman, E.F., Riley, S.C., Farha, S.A., **Maitland, B.M.**, Tucker, T.R., Provo, S.A., & McLean, M.W. (2013). *Status and Trends of the Lake Huron Offshore Demersal Fish Community, 1976-2012*. USGS Great Lakes Science Center.

Theses

2. **Maitland, B.M.**, (2020). *Isotopic Ecology of Aquatic Communities Along the Rocky Mountains— Great Plains Ecotone* [PhD thesis]. University of Wyoming.

1. **Maitland, B.M.**, (2015). *Stream Crossings in the Western Boreal Forest: Assessing Impacts and Prioritizing Restoration for Native Freshwater Fishes* [Master's thesis]. University of Alberta.

Pre-prints and in-review

2. Vagnon, C., Jardine, T.D., Hayden, B., Tunney, T.D., Sanchez-Hernandez, J., McIntosh, A.A., Alves, C.B., Arostegui, M.C., Baudry, T., Beesley, L., Belliard, J., Bellouard, N., Boulêtreau, S., Britton, R., **Maitland, B.M.**, ... Cucherousset, J. (under review). IsoFreshID: A global database of individual-level $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ stable isotopes of freshwater food webs. *Scientific Data*.
1. Sparks, M.M., Leavell, B.C., & **Maitland, B.M.**, (under review). A generalizable tool for predicting developmental phenology for wild poikilotherms. *Ecology and Evolution*.

Unpublished working papers

2. Hargrove, J., **Maitland, B.M.**, Sparks, Morgan M., & Isaak, D.J. (2026). *From thermal barriers to connectivity corridors: Shifting drivers of Rainbow Trout introgression into Westslope Cutthroat Trout over two decades*.
1. **Maitland, B.M.**, Thurow, R., Sparks, Morgan M., Isaak, D., & Buffington, J. (2026). *Habitat heterogeneity and phenotypic diversity: The influence of stream attributes on timing of Chinook Salmon spawning*. https://github.com/morgan-sparks/mfsr_phenology

Grants

Since 2015, I have led, co-led, or partnered on \$1.1 million in funded projects and contributed to multiple invited proposals exceeding \$2 million in scope. Funded and pending grants > \$10,000 are given below:

Pending	Lead-PI. "Idaho Bull Trout food web ecology and hybrid interactions". <i>Funding from U.S. Fish and Wildlife Service</i> . Collaborators: M. Sparks (Co-PI).	\$20,000
Pending	Co-PI. "Idaho Bull Trout migration timing and habitat use". <i>Funding from U.S. Fish and Wildlife Service</i> . Collaborators: M. Sparks (Lead-PI), B. Bowersox.	\$159,900
2026–2028	Co-PI. "From flames to fish: Co-management for wildfire and aquatic species in Idaho". <i>Funding from Joint Fire Science Program</i> . Collaborators: J. Peirce (Lead-PI), B. Sadegh, A. Bergstrom, C. Wasler, M. Sparks, D. Isaak.	\$150,284
2025–2029	Partner. "IRES: Climate change resilience in an aquatic bright spot: Lakes and rivers of Northern Mongolia". <i>Funding from NSF IRES</i> . Collaborators: O. Jensen (Lead-PI), G. Batsaikhan, H. Bradley, S. Chandra, C. Free, T. Goldberg, K. Hartman, P. Lopez-Duarte, B. Mendsaikhan, T. Miller, M. Provost, J. Weber, T. Young.	\$749,133
2025–2027	Lead-PI. "Monitoring Chinook Salmon spawning from space". <i>Funding from Trout Unlimited</i> . Collaborators: D. Daulwater, M. Sparks.	\$38,122
2024–2027	Lead-PI. "Chinook Salmon life history and otolith microchemistry". <i>Funding from UC-Davis and USDA Forest Service</i> . Collaborators: C. Jeffries.	\$50,184
2020–2021	Lead-PI. "Accessible underwater landscapes for non-STEM audiences". <i>Funding from UWyo Biodiversity Institute</i> . Collaborators: F. Rahel.	\$19,866
2019–2020	Lead-PI. "Diet specialization along stream gradients". <i>Funding from UWyo Zoology</i> . Collaborators: F. Rahel.	\$22,398
2015–2018	Lead-PI. "Factors that influence the diversity of fish assemblages in Wyoming". <i>Funding from UWyo Biodiversity Institute</i> . Collaborators: F. Rahel.	\$25,207

Invited full proposals that were withdrawn due to federal funding issues in the U.S.:

2025	Lead-PI. "A palaeoecological toolset to address shifting baseline syndrome". <i>Funding from DoD SERDP</i> . Collaborators: M. Sparks, B. Finney, K. Carim, A. Fraik, R. Thurow, C. Shobe, D. White.	\$1,500,000
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Invited full proposals that were not awarded:

2023	Lead-PI. "Cross-lake comparison of habitat coupling in Lakes Michigan and Superior". <i>Funding from WI Sea Grant</i> . Collaborators: O. Jensen.	\$258,571
2021	Lead-PI. "Climate change and groundwater pumping impacts on Wisconsin trout fisheries". <i>Funding from USGS MW CASC</i> . Collaborators: A. Latzka, O. Jensen, S. Loheide.	\$399,853

Honors and Awards

Professional awards

- 2026 Outstanding Service Award, U.S. Forest Service, Rocky Mountain Research Station
- 2024 Outstanding Service Award, U.S. Forest Service, Rocky Mountain Research Station
- 2021 Best Professional Presentation Award, Wisconsin Chapter American Fisheries Society

Fellowships

- 2020 Wisconsin Water Resources Postdoctoral Fellow, University of Wisconsin-Madison
- 2019 Dean's Graduate Fellowship, College of Arts and Sciences, University of Wyoming

Student Awards

- 2018 Eugene Maughan Graduate Award, Western Division American Fisheries Society
- 2018 Hank Gardner Graduate Award, Department of Zoology and Physiology University of Wyoming
- 2017 Clemens-Rigler Award, Canadian Aquatic Resources Section American Fisheries Society
- 2017 Board of Visitors Award, College of Arts and Sciences University of Wyoming
- 2017 Dennis Anderson Fisheries Award, Department Zoology and Physiology University of Wyoming
- 2017 William Trachtenberg Scholarship, Sustainable Fisheries Foundation
- 2017 Vern Bressler Fisheries Award, Department of Zoology and Physiology University of Wyoming
- 2015 Excellence in Fisheries Award, Canadian Aquatic Resources Section American Fisheries Society
- 2015 Clemens-Rigler Award, Canadian Aquatic Resources Section American Fisheries Society

Invited Academic Talks

- 2020-May "Exploring isotopic niches in Rocky Mountain Great Plains stream fish". Department of Zoology and Physiology, University of Wyoming.
- 2022-Mar "Coldwater fish in a changing world: Actionable ecology for management and policy". Department of Ecology, Montana State University.
- 2023-Dec "Freshwater science in a changing world: fisheries research for conservation and management". School of Freshwater Science, UW-Milwaukee.
- 2023-Apr "Fish and freshwater in a changing world: Science for management and conservation". Rocky Mountain Research Station, US Forest Service.
- 2025-Feb "Freshwater science for tomorrow: advancing fish ecology, conservation, and management". Wildlife Biology Program, University of Montana.
- 2026-Feb "Stream fish ecology and conservation under environmental change". Department of Fish, Wildlife, and Conservation Biology, Colorado State University.

Teaching

Instructor of Record

ZOO 4330 Ichthyology University of Wyoming Spring 2020 3 credits 22 students

Teaching Assistant and Lab Instructor

ZOO 4330	Ichthyology	University of Wyoming	Spring 2020	3 credits	22 students
ZOO 4330	Ichthyology	University of Wyoming	Spring 2019	3 credits	20 students
ZOO 4330	Ichthyology	University of Wyoming	Spring 2018	3 credits	25 students
ZOO 4330	Ichthyology	University of Wyoming	Spring 2017	3 credits	19 students
ZOO 4330	Ichthyology	University of Wyoming	Spring 2016	3 credits	18 students
ZOO 4310	Fisheries Management	University of Wyoming	Fall 2019	3 credits	21 students
ZOO 4310	Fisheries Management	University of Wyoming	Fall 2018	3 credits	26 students
ZOO 4310	Fisheries Management	University of Wyoming	Fall 2017	3 credits	22 students
ZOO 4310	Fisheries Management	University of Wyoming	Fall 2017	3 credits	19 students
ZOO 4310	Fisheries Management	University of Wyoming	Fall 2017	3 credits	23 students
ZOO 4310	Fisheries and Wildlife Management	University of Alberta	Spring 2015	4 credits	27 students
ZOO 4310	Fisheries and Wildlife Management	University of Alberta	Spring 2014	4 credits	29 students

Guest Lectures

- 2023-Feb "Stable isotope mixing models in R". RENR 5500, University of Wyoming.
- 2020-Dec "Aquatic ecology in Western ecosystems". LIFE 3400, University of Wyoming.
- 2019-Apr "Applications of stable isotopes in fisheries management". ZOO 4310, University of Wyoming.

Mentoring

Post-baccalaureate

2025–Present	Garrison Ferone	University of Idaho
2023–2025	Dylan Keisser	University of Tennessee, Chattanooga

Undergraduate

2024–2025	Trevor Davis	University of Tennessee, Chattanooga
2023–2024	Jonathan Martin	University of Wisconsin-Madison
2018–2020	Nathan Barrus	University of Wyoming
2019–2020	Russell Brubaker	University of Wyoming
2018–2019	Reilly Davis	University of Wyoming
2017	Brynn Hirschman	University of Wyoming
2017	Austin Jacobson	University of Wyoming
2016	Shaun Helsper	University of Wyoming
2016–2017	Travis Allison	University of Wyoming
2016	Andrew Annear	University of Wyoming
2015	Jake Werner	University of Wyoming

Professional Service and Leadership

Service

2022–2024	Field Operation Facilitation and Boat Committee, Center for Limnology, UW-Madison
2021–2022	Member, Fisheries Working Group, Wisconsin Initiative on Climate Change Impacts
2020–2022	Member, Trout Species Team, Wisconsin Department of Natural Resources
2019–2020	Faculty Search Committee, Department of Zoology and Physiology, University of Wyoming
2016–2020	Social Media Coordinator, Program in Ecology, University of Wyoming
2016–2018	Excom Member, CO-WY State Chapter, American Fisheries Society
2016–2017	Graduate Council, student representative, University of Wyoming
2016–2017	President, Program in Ecology, University of Wyoming
2016–2017	President, UWyo Student Chapter, American Fisheries Society
2015–Present	Member, Freshwater Working Group, Society for Conservation Biology

Society memberships

2019–Present	Galápagos Conservancy
2016–Present	Society for Freshwater Science
2016–2019	Association for the Science of Limnology and Oceanography
2015–Present	Ecological Society of America
2010–Present	American Fisheries Society
2010–Present	Society for Conservation Biology
2010–2016	International Association of Great Lakes Research

Society service

2026	Program co-chair, Fish and Fire, American Fisheries Society Annual Meeting
2024	Program co-chair, Science to Restore and Manage Wildland Aquatic Habitats and Watersheds, AGU Fall Meeting
2016	Program co-chair, Western Division American Fisheries Society Student Colloquium

Reviewer

- Proposal reviewer
 - Lake Champlain Fisheries Research Program
 - NSF, Division of Environmental Biology (DEB)
- Manuscript reviewer:
 - Since 2016, I have reviewed 55 manuscripts from 18 journals (~ 5.5 per year).
 - Journals include: Fisheries Research, Oecologia, Global Change Biology, Ecological Applications, Ecosphere, Freshwater Biology, Freshwater Science, Fisheries, North American Journal of Fisheries Management, Frontiers in Ecology and Evolution, Aquatic Invasions, Hydrobiologia, Journal of Ecology, Environmental Biology of Fishes, Restoration Ecology, PeerJ, River Research and Applications, Journal of Great Lakes Research Diversity and Distributions, Transportation Research: Part D.

Leadership development

2021	Equity and Inclusion Workshop	EQT By Design via WI Sea Grant
2008	Leave No Trace Master Educator	Center for Outdoor Ethics, NOLS

Outreach and Broader Impacts

Hosted workshops

- 2023–Feb “Programming with R”. Wisconsin Chapter AFS Annual Meeting, Stevens Point, WI.
 2022–Feb “Data vizualization in R”. Wisconsin Chapter AFS Annual Meeting, Virtual.

Media and public talks

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| 2023–Feb | Milwaukee Journal Sentinel | Topic: Climate change and stream trout |
| 2023–Jan | Live Interview: WPR Morning Show | Topic: Climate change and stream trout |
| 2022–Dec | Water Blogged | Topic: Climate change and stream trout |
| 2022–Mar | Trout Unlimited, Madison Chapter | Topic: Climate change and stream trout |
| 2017–Nov | Magazine Article: The Tributary | Topic: Cutthroat trout |
| 2017–Aug | U Wyoming Science Cafe | Topic: Aquatic connectivity |
| 2016–Jun | U Wyoming Science Cafe | Topic: Aquatic connectivity |

Public service and outreach

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| 2022–2023 | Food service volunteer | Ronald McDonald House, Madison, WI |
| 2016–2020 | Science outreach and stream ecology field trips | Laramie High School, Laramie, WY |
| 2016–2019 | Group leader, Wyoming Bio-Blitz | Biodiversity Institute, University of Wyoming |
| 2016–2018 | Big Brother | Big Brothers Big Sisters of Wyoming |
| 2017–2018 | Habitat restoration projects | Girl Scouts of America |
| 2016–2017 | Science judge | Wyoming State Science Fair |

Conference Presentations

Underlined names indicate undergraduate research mentees.

47. Valentine, G.P., Shobe, C.M., Alfred, V.S., Lininger, K.B., Meyer, K., & **Maitland, B.M.**, (2026, May). *Two years of benthic macroinvertebrate community recovery following low-head dam removal in a semi-arid rocky mountain stream* [Paper]. SFS Annual Meeting, Spokane, WA.
46. Walker, R.H., Daugherty, O., Davis, T., Wildman, D., Wright, L., Hunt, Darin, & **Maitland, B.M.**, (2026a, May). *That’s a mouthful: Availability, feeding ecology, and life history variation of stream fishes during periodical cicada emergences* [Paper]. SFS Annual Meeting, Spokane, WA.
45. Wildman, D., Daugherty, O., Davis, T., Wright, L., Hunt, Darin, **Maitland, B.M.**, & Walker, R.H. (2026, May). *Cicadas in the stream: Quantifying periodical cicada drift densities in streams* [Poster]. SFS Annual Meeting, Spokane, WA.
44. Thurow, R.F., **Maitland, B.M.**, Sparks, M.M., Isaak, D.J., & Buffington, J.M. (2026, April). *Thermal heterogeneity structures diverse chinook salmon spawning portfolios across a stream network* [Paper]. WD AFS Annual Meeting, Portland, OR.
43. Walker, R.H., Daugherty, O., Davis, T., Wildman, D., Wright, L., Hunt, Darin, & **Maitland, B.M.**, (2026b). *Cicadas in the stream: Evaluating the foraging ecology and life history variation of fishes during a rare but massive resource pulse to stream ecosystems* [Paper]. TN Chapter AFS Annual Meeting, Cumberland Mountain State Park, TN.
42. Ferone, G.S., **Maitland, B.M.**, Isaak, D.J., & Caudill, C.C. (2025, November). *Applying spatial stream networks and species distribution models to two decades of Idaho DEQ BURP stream benthic macroinvertebrate samples* [Paper]. SFS Pacific Northwest Chapter Annual Meeting, Boise, ID.
41. Sparks, M.M., Leavell, B.C., & **Maitland, B.M.**, (2025, November). *A generalizable tool for predicting developmental phenology for freshwater poikilotherms* [Paper]. SFS Pacific Northwest Chapter Annual Meeting, Boise, ID.
40. Kirk, M.A., & **Maitland, B.M.**, (2025, August). *Trophic overlap between invasive round gobies and native benthic darters in a stream biodiversity hotspot* [Paper]. AFS annual meeting, San Antonio, TX.
39. **Maitland, B.M.**, Sparks, M.M., Felts, E., Swartz, A., & Frater, P. (2025a, August). *hatchR: A toolset for predicting when fish hatch and emerge* [Poster]. AFS annual meeting, San Antonio, TX.
38. **Maitland, B.M.**, Sparks, M.M., Felts, E., Swartz, A., & Frater, P. (2025b, May). *hatchR: A toolset for predicting when fish hatch and emerge* [Poster]. Western division AFS annual meeting, Westminster, CO.
37. **Maitland, B.M.**, Sparks, M.M., Felts, E., Swartz, A., & Frater, P. (2025c, April). *hatchR: A toolset for predicting when fish hatch and emerge* [Paper]. Idaho chapter AFS annual meeting, Boise, ID.
36. Davis, T., Wright, L., Keisser, D., **Maitland, B.M.**, & Walker, R.H. (2024, November). *That’s a mouthful: Periodical cicadas, fish body size, and feeding ecology* [Poster]. Southeastern fisheries council meeting, Little Rock, AR.
35. Keisser, D., **Maitland, B.M.**, & Walker, R.H. (2024, November). *Quantifying rare allochthonous insect pulses: How many periodical cicadas enter streams?* [Paper]. Southeastern fisheries council meeting, Little Rock, AR.
34. Walker, R.H., Keisser, D., **Maitland, B.M.**, & Hunt, D. (2024, November). *Quantifying the availability, use, and consequences of periodical cicadas in stream food webs* [Paper]. Southeastern fisheries council meeting, Little Rock, AR.
33. Hauxwell, J.A., Voter, C.B., Guerrero-Bolaño, F.J., Latzka, A.W., & **Maitland, B.M.**, (2024, June). *Adaptable university-agency early-career fellowship program creates win-win-win for Wisconsin’s Waters* [Paper]. IAGLR annual conference, Milwaukee, WI.

32. **Maitland, B.M.**, (2024, June). *Testing food web theory in a large lake: The role of body size in habitat coupling in Lake Michigan* [Paper]. ASLO annual meeting, Madison, WI.
31. **Maitland, B.M.**, & Latzka, A.W. (2024, May). *Shifting climate conditions affect recruitment in Midwestern trout, but depend on seasonal and spatial context* [Paper]. Washington, british colombia, and idaho chapters AFS joint meeting, Spokane, WA.
30. Walker, R.H., & **Maitland, B.M.**, (2024, May). *The cicadian rhythm: Preparation for evaluating the availability, use, and consequences of periodical cicadas to stream fishes* [Paper]. Iowa chapter AFS annual meeting, Iowa City, IA.
29. **Maitland, B.M.**, Jensen, O.P., Hoffman, J.C., & Keppeler, F.W. (2023a, August). *Testing food web theory in large lakes: The role of body size in habitat coupling in Lake Michigan* [Paper]. AFS annual meeting, Grand Rapids, MI.
28. Clancy, N., Isaak, D., **Maitland, B.M.**, Budy, P., & Walters, A. (2023, March). *Thermal refuges for climate vulnerable fishes in Wyoming* [Paper]. Colorado/Wyoming chapter of the American Fisheries Society, Laramie, WY.
27. **Maitland, B.M.**, Jensen, O.P., Hoffman, J.C., Keppeler, F.W., Bunnell, D.B., McNaught, A.S., & Gerig, B.S. (2023b, February). *Body size, trophic position, and energy coupling in Lake Michigan* [Paper]. Wisconsin chapter AFS annual meeting, Stevens Point, WI.
26. Sass, G.G., Beard, D., Broda, S., Carpenter, S.R., Chipps, S.R., Cichosz, T.A., Dassow, C.J., Elwer, B., Embke, H.S., Feiner, Z.S., Gaeta, J.W., Hansen, G.J.A., Hansen, J., Hennessy, J., Homola, J.J., ... Zebro, L. (2023, January). *Depensatory recruitment in Wisconsin walleye: Mechanisms and management* [Paper]. Wisconsin DNR office of applied science webinar series, Virtual.
25. Ward, N.K., Lynch, A.J., Beever, E.A., Bouska, K.L., Embke, H.S., Kocik, J.F., Limpinsel, D., Magee, M.R., **Maitland, B.M.**, Morton, J.M., Mueller, J.M., Oliver, D.C., Rantala, H.M., Sass, G.G., Schultz, A., ... Wilkening, J.L. (2022, August). *Using the Resist-Accept-Direct (RAD) Framework to Reimagine Large River Management* [Paper]. AFS annual meeting, Spokane, WA.
24. **Maitland, B.M.**, & Latzka, A.W. (2022, May). *Shifting climate conditions affect recruitment in Midwestern stream trout but depend on seasonal and spatial context* [Paper]. Joint aquatic sciences meeting, Grand Rapids, MI.
23. **Maitland, B.M.**, & Latzka, A.W. (2021, November). *Long-term trends of Midwestern stream trout populations in a changing climate, and the effects of seasonal weather* [Paper]. AFS annual meeting, Baltimore, MD.
22. **Maitland, B.M.**, Latzka, A.W., Frater, P., & Mitro, M.G. (2021a, May). *Untangling the effects of hydrologic change on trout in Wisconsin streams* [Paper]. Wisconsin water week, Virtual.
21. Lapidés, D.A., **Maitland, B.M.**, & Pruitt, A.H. (2021, March). *Assessing approaches to quantify hydrological alteration on Wisconsin's streams* [Paper]. Wisconsin section AWRA annual meeting, Virtual.
20. **Maitland, B.M.**, Voter, C.B., Guerrero-Bolaño, F.J., Latzka, A.W., & Hauxwell, J.A. (2021b, March). *Tackling Wisconsin's water challenges through UW Water Science-Policy Fellowships and agency partnerships* [Paper]. Wisconsin water week, Virtual.
19. **Maitland, B.M.**, Latzka, A.W., Frater, P., & Mitro, M.G. (2021c, February). *Long-term trout trends in Wisconsin and the role of hydrologic change* [Paper]. Wisconsin chapter AFS annual meeting, Virtual.
18. **Maitland, B.M.**, Fetzter, W.W., Collins, S.M., & Rahel, F.J. (2020, March). *Non-lethal approaches for obtaining stable isotope data for diet studies of North American fishes* [Paper]. CO/WY chapter AFS annual meeting, Laramie, WY.
17. Barrus, N.T., **Maitland, B.M.**, & Rahel, F.J. (2019, October). *Identification of an ideal baseline organism for stream food web studies* [Paper]. Joint meeting of the wildlife and american fisheries society, Reno, NV.
16. **Maitland, B.M.**, & Rahel, F.J. (2019, October). *Stream food web expansion in response to changing environmental conditions* [Paper]. Joint meeting of the wildlife and american fisheries society, Reno, NV.
15. **Maitland, B.M.**, & Rahel, F.J. (2018, January). *Niche partitioning in fish assemblages along longitudinal stream gradient* [Poster]. Canadian conference for freshwater fisheries research, Edmonton, AB.
14. **Maitland, B.M.**, & Rahel, F.J. (2017, June). *Isotopic niches and the longitudinal gradient in Rocky Mountain-Great Plains streams* [Paper]. SFS annual meeting, Raleigh, NC.
13. **Maitland, B.M.**, Walker, R.H., LaSharr, T.N., Rosin, M., & Ben-David, M. (2017a, April). *Juvenile salmonids stranded in flood ponds of Alaska* [Paper]. Program in ecology symposium, Laramie, WY.
12. **Maitland, B.M.**, Walker, R.H., LaSharr, T.N., Rosin, M., & Ben-David, M. (2017b, February). *The fate of juvenile salmonids stranded in flood ponds: Implications for nutrient transfers* [Paper]. Utah and colorado/wyoming chapter AFS joint meeting, Grand Junction, CO.
11. **Maitland, B.M.**, Poesch, M.S., & Anderson, A.E. (2016a, May). *Culverts in forested watersheds: Impacts and restoration methods for native freshwater fishes* [Paper]. SFS annual meeting, Sacramento, CA.
10. Annear, A.A., **Maitland, B.M.**, & Rahel, F.J. (2016, March). *Stable isotope quantification of resource use by crayfish in the Laramie River to inform lotic food web analyses* [Poster]. Colorado/wyoming chapter AFS annual meeting, Laramie, WY.
9. **Maitland, B.M.**, Poesch, M.S., & Anderson, A.E. (2016b, March). *Culverts in forested watersheds: Impacts and restoration methods for native freshwater fishes* [Paper]. Colorado/wyoming chapter AFS annual meeting, Laramie, WY.
8. Daniels, S.E., Fitzpatrick, T.A., & **Maitland, B.M.**, (2015, January). *Idiosyncratic resource use by Laramie raccoons: Are trash pandas, trash pandas?* [Poster]. Stable isotope ecology symposium, Laramie, WY.

7. Hirschfield, F., Anderson, A.E., **Maitland, B.M.**, Wagner, M., Zaichkowsky, M., & Mottishaw, C. (2014, October). *Integrating science and management to improve sustainability of forested watersheds in Northwestern Canada* [Poster]. SAF national convention, Salt Lake City, UT.
6. **Maitland, B.M.**, Poesch, M.S., & Anderson, A.E. (2014a, August). *Stream crossing impacts on freshwater fishes: Implications for management* [Poster]. Forest industry lecture series, Edmonton, AB.
5. **Maitland, B.M.**, O'Malley, B.P., O'Brien, T.P., Armenio, P.M., Watson, N.M., & Roseman, E.F. (2014b, May). *Zooplankton community dynamics in a northern Lake Huron embayment: The influence of water temperature and larval fish abundance* [Paper]. IAGLR annual meeting, Hamilton, ON.
4. **Maitland, B.M.**, Poesch, M.S., & Anderson, A.E. (2014c, May). *Stream crossing assessment procedures as a tool for mitigating impacts on freshwater fish* [Poster]. Columbia mountain institute of applied ecology's resource roads conference, Nelson, BC.
3. **Maitland, B.M.**, Poesch, M.S., & Anderson, A.E. (2014d, January). *Stream crossing impacts on freshwater fish and in-stream habitat in Alberta* [Paper]. Canadian conference for freshwater fisheries research, Yellowknife, NWT.
2. **Maitland, B.M.**, Farha, S.A., Darnton, R.W., Smith, K.R., & Riley, S.C. (2012, September). *2012 Avian botulism environmental sampling in Sleeping Bear Dunes National Lakeshore* [Paper]. Workshop on GLRI project no. 73 and 91 USGS/NPS meeting, Empire, MI.
1. Discenza, J.D., O'Malley, B.P., **Maitland, B.M.**, & Fernando, D. (2011, April). *Analyzing the genetic variation in American Hart's tongue fern* [Poster]. SUNY: ESF spotlight on student's research, Syracuse, NY.